

Preliminary Area Summary & Research Question Framework

Domain

The broad field I'm exploring is AI security and Large Language Model (LLM) safety, with a focus on adversarial prompt attacks, jailbreak detection, and trustworthy AI. The specific problem area centers on how obfuscated jailbreak prompts and prompt injection attacks bypass existing LLM safety guardrails, and whether transformer-based deep learning models can reliably detect these malicious inputs before they compromise model behavior. This intersects cybersecurity, machine learning security, and natural language processing.

Horizon Scan Observations

After scanning 8 research papers on this topic, I identified three key observations:

What is being studied extensively: Detection methods for prompt injection attacks are receiving significant attention using transformer-based approaches. The "Attention Tracker" paper proposes tracking attention patterns to identify the "distraction effect" where important attention heads shift focus from original instructions to injected instructions, achieving AUROC improvements up to 10% over existing methods without requiring additional training. Multiple classifier-based approaches (BERT, LSTM, Random Forest) are also being trained on curated prompt injection datasets from sources like HackAPrompt. Comprehensive surveys provide detailed taxonomies dividing attacks into black-box/white-box and defenses into prompt-level/model-level categories.

Where there seems to be disagreement: There is tension between detection effectiveness and computational cost. While Attention Tracker claims to be training-free with minimal computational overhead, other research suggests that specifically trained detection models (like BERT-based classifiers) show satisfactory performance whereas existing LLM-based detectors (Llama-Guard, Protect-AI, Meta's Prompt-Guard) struggle with indirect prompt injection, achieving only 39-86% accuracy depending on attack type. The "LLM Jailbreak Detection for (Almost) Free!" paper argues that detection can work with nearly zero computational cost using output distribution differences, contradicting claims that heavy inference or trained models are necessary.

What appears to be missing or underexplored: Obfuscated activation attacks that specifically bypass latent-space defenses are critically underexplored. Research demonstrates that state-of-the-art latent-space monitors (sparse autoencoders, representation probing, latent OOD detection) can have recall reduced from 100% to 0% while maintaining 90% jailbreaking success when attackers use obfuscation techniques. There's also a significant gap in detection methods for attacks that omit explicit instructions or employ imperceptible perturbations—detectors largely fail against these more sophisticated obfuscated attacks. Additionally, benchmarks specifically testing transformer-based deep learning models (BERT, DistilBERT) against obfuscated jailbreak prompts rather than direct prompt injection are limited.

Research Question Framework

Framework Used: SPIDER (Sample, Perspective, Interval, Design, Evaluation)

Completed SPIDER Template:

- **Sample:** Large Language Models (Llama 3, Gemma 2, Mistral, DeepSeek-R1, Qwen variants, Phi)
- **Perspective:** Security researchers and AI safety engineers developing transformer-based detection systems
- **Interval:** Inference-time (real-time detection before model response generation)
- **Design:** Training and evaluating transformer-based deep learning models (BERT, DistilBERT) on obfuscated jailbreak and prompt injection datasets

- **Evaluation:** Detection accuracy (AUROC, precision, recall, F1-score), computational cost (inference time, memory), generalization across LLM architectures and attack types

Finalised Research Question: Can transformer-based deep learning models accurately detect obfuscated jailbreak and prompt injection attacks in Large Language Models while maintaining efficient inference-time performance?

Why This Matters

This question is worth investigating because prompt injection and jailbreak attacks can compromise security, privacy, and functionality of LLM-integrated systems—31 out of 36 real LLM applications were found susceptible to prompt injection attacks, potentially impacting millions of users. Transformer-based detection offers a practical balance between accuracy and computational efficiency compared to both LLM-based detectors (which are expensive) and traditional classifiers (which may lack robustness), making it viable for real-time deployment in production systems.